

Geodesica

An introduction to non-Euclidean geometry

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1 Introduction

What is non-Euclidean geometry? I believe most people in the modern world have a vague understanding of what non-Euclidean means. It is often connotated with a reality that isn't like ours and it is often used as an umbrella term to describe all sorts of related concepts. In this handout we try to give a mathematical introduction to the concepts of non-Euclidean geometries. As a visual aide throughout this journey we make use of tessellations.

This document serves as a self-study guide that picks up on topics from discrete mathematics and linear algebra. The reader should be familiar and comfortable handling graph theory, complex numbers and linear transformations. It is worth looking up the following concepts as these will serve as our basis.

- Euler Characteristic $\chi = V - E + F$
- Euler Formula $e^{i\phi} = \cos(\phi) + i \sin(\phi)$
- Polar coordinates (θ, r)
- Spherical coordinates (θ, ϕ, r)
- Orthogonal Projection
- Stereographic Projection

We highly encourage you to read further into the matter if you are interested. This handout merely scrapes the top of this vast field of mathematics.

1.1 Historical Background

In order for us to get a grasp on non-Euclidean geometries, we should first understand what makes a geometry Euclidean in the first place. We should all be relatively familiar with Euclidean geometry, it is what most of us understand as **the** theory of geometry and what we are taught out of high school. Euclid of Alexandria, our namesake, formulated the ancient Greek's understanding of geometry, which later formed the classical understanding of geometry. In his work Euclid gives us the definition of geometric elements, which coincide with our modern understanding. Furthermore Euclid states five Axioms.

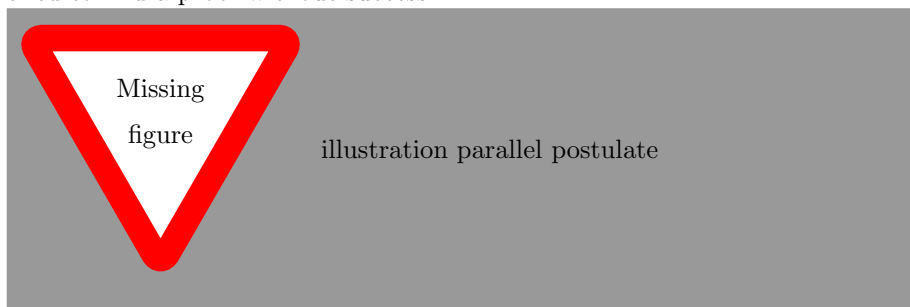
1. To draw a straight line [segment] from any point to any point.
2. To produce a finite straight line [segment] continuously in a straight line.
3. To describe a circle with any center and distance.
4. That all right angles are equal to one another.
5. That, if a straight line falling on two straight lines make the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than two right angles.[pp. Euclid's elements]

The first four axioms are very easily interpretable. Two points can be connected by a straight line segment, line segments can be infinitely extended into

mention
notable
mathemati-
cians in
this field?
see (see
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lines, there exists a circle with any center and radius and all the right angles must be equal to one another.

The fifth postulate, also coined the parallel postulate, is where things get interesting. At first glance it seems out of place. A simpler but equivalent formulation this is given by Playfair's axiom. It states that on a plane, given a line and a point not on it, only one parallel line to the given line can be drawn through the point. Throughout the ages mathematicians have speculated that the fifth axiom is superfluous and could be derived from the first four. Many tried to find a proof without success.



A geometry becomes non-Euclidean if we modify the fifth axiom (**cite?**). There are two non-Euclidean geometries that arise from this change. Elliptic Geometry, where there are exist zero parallels to a given one and hyperbolic Geometry where there exist infinitely many.

1.2 Topology

1.3 Gaussian-Curvature

When discussing non-Euclidean Geometry one will sooner or later fall upon the term gaussian-curvature. Gaussian curvature is a measure of how much a 2-dimensional surface curves at a specific point in space.

We call a curvature constant if it is the same at each point.

Gaussian-Curvature is an intrinsic property of a surface. Meaning that a being living entirely on the surface could determine it's measure.

The sphere gives us a surface with constant positive curvature. We call this surfaces embedded in $\mathbb{R}^3$

There exist surfaces with negative curvature in $\mathbb{R}^3$, such as the hyperboloid and paraboloid.

However Hilbert proofed that there are no embeddings of a surface with constant negative curvature in 3-d space. This is the main reason why it is so hard for us to imagine hyperbolic space.

1.4 Tessellations

A *tessellation* (or Tiling) is the process of covering a surface with geometrical shapes so that there are no gaps or overlaps. The shapes used in a tiling are

When omitting the fifth axiom completely we talk about absolute geometry Faber 1983, pg. 131,

State Hilbert's axioms, or mention it as the modern axiomatization, see handout from uni-bielefeld

try to give analog from 1d curvature

show paraboloid and hyperboloid

called *tiles*. Tilings are classified by restricting the choice and alignment of tiles.

1.4.1 Polygonal Tessellations

A *polygonal* tessellation is a tessellation where we restrict the tiles to be polygons. If we further restrict the alignment of tiles, such that adjacent tiles share a common edge and meet at their vertices, we talk about an *edge-to-edge* tiling.

Polygonal edge-to-edge tilings can be classified further using the following properties.

A tiling is said to be *regular*, if all it's tiles are regular polygons.

A tiling is said to be *face*-transitive, if all it's tiles are congruent.

A tiling is said to be *edge*-transitive, if

1.4.2 Triangle Tiling

2 Tessellations of the euclidean plane

2.1 Uniform Tilings

Uniform tilings are a larger

Definition 1 (Vertex transitive) *A tiling is said to be vertex transitive if all of its vertices are indistinguishable from each other*

- ☐ (playful) take some regular polygons, find arrangements to cover the plane without gapping
- ☐ what is a tessellation
- ☐ what makes it regular

2.2 Tilings of the plane

- ☐ definition face, edge, vertex transitivity

2.3 Regular Tessellations

- ☐ platonic tilings
- ☐ proof that there are only three
- ☐ Schläfli Symbol $\{p, q\}$ describes a tiling where q p -gons meet at each vertex

2.4 Semi- and Quasi-Regular Tessellations

- ☐ archimedean tilings
 - ☐ most are semi-regular derived by truncation
 - ☐ 3.6.3.6 is the only quasi-regular archimedean tiling
- ☐ in a later chapter we can derive semi-regular tilings by truncating, and quasi-regular tilings by rectifying regular tilings.

https://en.wikipedia.org/wiki/Euclidean_tilings_by_convex_regular_polygons#Archimedean,_uniform_or_semi-regular_tilings

2.5 Quasi-Regular Tessellations

- ☐ in a later chapter we can derive quasi regular tilings by rectifying

2.6 Tessellation with Möbius Triangles

- ☐ requires discussion about angle sum

A triangle tiling is a tiling made from Schwarz triangles.

A *Möbius triangle* is a triangle defined by the three tuple $(pqr) \in \mathbb{Z}^3$, which represent it's internal angles. The angle at each vertex are given by

$$\alpha = \frac{\pi}{p} \quad \beta = \frac{\pi}{q} \quad \gamma = \frac{\pi}{r}$$

Not all choices for (pqr) yield a valid triangle. Must ensure that the angles form a valid internal angle sum. For a triangle in the Euclidean Plane the angles must add up to π , hence (p, q, r) define a Euclidean triangle exactly when

$$\alpha + \beta + \gamma = \pi \Rightarrow \frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$$

We can tessellate a surface by continuously reflecting a Möbius triangle along its edges.

Note that all permutations of p, q, r will produce a similar triangle and henceforth congruent tilings, $(p, q, r) \cong (q, r, p) \cong (r, p, q)$, etc.

2.6.1 Proof of Existence

Let $p, q, r \in \mathbb{Z}^+$ be the coefficients of a Möbius triangle. We shall proof for which choices of p, q, r there exists a valid **Euclidean** tiling. In order to formulate an exhaustive proof, we order the coefficients such that $p \leq q \leq r$. This way we will cover all unique tilings without repetition.

Our strategy in this proof is to consecutively limit the number of choices for each coefficient and make a case by case analysis for each one.

In Euclidean geometry the angle sum of a triangle is π , therefore $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$ {#eq:test}.

We want p to assume the its lowest possible values, i.e. to represent the largest possible angle in our triangle. As a lower bound we have $2 \leq p$, since if p was one, the other coefficients would have to be 0. Our upper bound is reached when $p = q = r$, therefore if we apply (eq:test?) we get $p \leq 3$.

We have now have found our values for $p \in \{2, 3\}$.

Note

2.7 Euler Characteristic

The Euler characteristic of a cell division is given by [Weeks, pp.171] Where V is the number of vertices, E the number of edges and F the number of faces

$$\chi = V - E + F$$

For polyhedra $\chi = 2$ since they can be embedded on the sphere $\mathbb{S}^2$ [Coxeter, pp. 10]

The Gauss-Bonnet formula gives the area of different surfaces

$$A_{\text{sphere}} = 2\pi\chi$$

3 Tessellations of the sphere

To begin our journey into non-euclidean geometries we will first look at geometry of the sphere. Understanding spherical geometry will give us helpful insights to understand and interpret hyperbolic geometry.

3.1 Geometry on the sphere

- ☐ Geodesics (what is the shortest path on the sphere)
- ☐ Antipodal
- ☐ Dihedral Angle

3.1.1 Great Circles

- ☐ Todo, definition for the unit sphere S and representation of planes by unit normal vectors $\hat{n}$

A great circle is given by the intersection of the sphere with a plane through it's center. For any two points on the great circle the arc formed between them is the shortest path the sphere.

- ☐ Show that this intersection is actually a circle?

Let A, B be two points on the sphere. The normal vector of a great circle through A and B is given by the cross product $\vec{A} \times \vec{B}$.

- ☐ We proof that arcs of great circles form the shortest path between two points
- ☐ Note that normal vectors on the xy-plane form great circles through the north and south pole of the sphere.
- ☐ The perpendicular line to a great circle through it's center is called it's axis
- ☐ The two point intersection of a great circles axis with the sphere are called it's poles.
- ☐ A great circle splits the sphere into two hemispheres

Let $x \in S$ be the elements of a sphere and $\hat{n}$ the unit normal vector of a great circle. The great circle partitions the sphere into two hemispheres:

- A positive hemisphere $S_+ = \{x \in S : x \cdot \hat{n} > 0\}$,
- a negative hemisphere $S_- = \{x \in S : x \cdot \hat{n} < 0\}$,

3.1.2 Spherical Lunes

- ☐ Special case of a digon
- ☐ Polygon with two vertices (possible in S, degenerate in E)

Let g_1, g_2 be two great circles and n_1, n_2 their respective unit normal vectors. The two circles intersect at an angle θ given by $\cos(\theta) = \hat{n}_1 \cdot \hat{n}_2$. Let S_{1+}, S_{2+} be the positive hemisphere given by the great circles. We define a spherical lune L to be the wedge shaped intersection of the two hemispheres $L = S_{1+} \cap S_{2+}$.

The area of a lune is given by it's angle $A(L_\theta) = 2\theta r^2$. A proof is left as an exercise to the reader :) This fact will be useful later.

Note that the intersection $L' = S_{1-} \cap S_{2-}$ is a congruent antipodal lune to L . Meaning $A(L_\theta) = A(L'_\theta)$.

□ Not sure if we should add this. It is rigorous but also obfuscating the point a lot

□ $\bar{L} = \overline{S_{1+}} \cap \overline{S_{2+}}$ lune including boundary

□ $A(L) = A(\bar{L})$ including boundary does not change area

Spherical Triangles Let $A, B, C \in S$ be points on the unit sphere forming the vertices of a spherical triangle. It's edges a, b, c are given by great circles with respective unit normal vectors $\hat{n}_a, \hat{n}_b, \hat{n}_c$ with

$$n_a = C \times B$$

$$n_b = A \times C$$

$$n_c = B \times A$$

It is good to be rigorous at this stage with how we define the great circles that form our triangle. We want $\Delta = S_{a+} \cap S_{b+} \cap S_{c+}$. This will be useful once we get into algorithms to generate tilings. Note this also has to be considered in Euclidean geometry when we represent lines in their normal form.

$$\alpha = n_b \cdot n_c$$

$$\beta = n_c \cdot n_b$$

$$\gamma = n_a \cdot n_b$$

The angle sum of a spherical triangle is greater than π . There is an elegant proof to show this.

The unit normal vectors are non-collinear and non-planar if and only if A, B, C span a triangle. This is exactly the case when the scalar triple product $[n_a, n_b, n_c] = n_a \cdot (n_b \times n_c) > 0$.

Let us construct the spherical lunes $L_\alpha, L_\beta, L_\gamma$ and their antipodal counterparts $L'_\alpha, L'_\beta, L'_\gamma$. Note that each lune contains either the original triangle Δ or it's antipodal counterpart Δ' . Note further that we can reconstruct the sphere by forming the unions of all lunes $\bigcup \bar{L} = S$.

Summing up the area of all lunes we get back the area of the sphere, however overcounting by four areas of our original triangle. Note our lemma $A(L) = A(L')$ hence $A(L) + A(L') = 2A(L)$. Simplifying we get an equation for the angle sum.

$$\begin{aligned}
2A(L_\alpha) + 2A(L_\beta) + 2A(L_\gamma) &= 4\pi r^2 + 4A(\triangle) \\
&\Downarrow \\
A(L_\alpha) + A(L_\beta) + A(L_\gamma) &= 2\pi r^2 + 2A(\triangle) \\
&\Downarrow \\
2\alpha r^2 + 2\beta r^2 + 2\gamma r^2 &= 2\pi r^2 + 2A(\triangle) \\
&\Downarrow \\
\alpha + \beta + \gamma &= \pi + \frac{A(\triangle)}{r^2}
\end{aligned}$$

This gives us a nice equation for the angle sum in relation to the triangles area. Let us consider the area bounds of a spherical triangle. It is clear that the area of a triangle must be greater than 0 and cannot exceed 2π . Therefore $0 < T < 2\pi$

Further we get a bound on the angle sum $\pi < \alpha + \beta + \gamma < 3\pi$.

This also gives us a formula for the area of a spherical triangle where

$$A(\triangle) = \alpha + \beta + \gamma - \pi$$

3.2 Constructing regular tessellations

3.2.1 Hosohedron

3.2.2 Triangle Tilings

- ☐ split hosohedron along equator
- ☐ combinations for pqr with angle sum constraint

3.2.3 Regular and Uniform Tessellations

- ☐ Dual graphs on triangle tilings
 - ☐ already getting into kaleidoscope construction through symmetry groups?

3.3 Excursion Polyhedra

- ☐ Tilings of the sphere give us graphs of 3d polyhedra sharing same symmetries and duality
 - ☐ Platonic solids
 - ☐ Prisms

4 Stereographic Projection

Now that we have an understanding of how lines and triangles behave on a sphere

As of now we have studied two-dimensional spherical geometry embedded on the surface of a three-dimensional object, the sphere. This is obviously very natural, yet it begs the question whether we could somehow flatten the surface of the sphere.

We already use projections for our figures, where the sphere is orthographically or perspective projected onto the 2d image. Which our brains are very used to seeing, we are able to reconstruct a 3d interpretation in our minds eye.

The questions is whether there are other projections that might prove more useful in comparing spherical geometry to euclidean and hyperbolic geometry. We want our projection to have useful properties.

- The projection should be bijective in order for us to have one to one correspondance between points in the projection and points in the embedding.
- it should preserve “useful” geometric properties
 - conformal (angle preserving)
 - geodesics should map into ”intuitive” counterpart
- it should come with a metric

Let $\Phi : S \subset \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the projection from the sphere to the plane and $\Phi^{-1} : \mathbb{R}^2 \rightarrow S \subset \mathbb{R}^3$ the unprojection that from the plane to the spehere. Let $(x, y, z) \in S$ denote points on the sphere and $(X, Y) \in \mathbb{R}^2$ points on the plane

$$\Phi(x, y, z) = \left(\frac{x}{1-z}, \frac{y}{1-z} \right) = (X, Y)$$

$$\Phi^{-1}(X, Y) = \left(\frac{2X}{X^2 + Y^2 + 1}, \frac{2Y}{X^2 + Y^2 + 1}, \frac{X^2 + Y^2 - 1}{X^2 + Y^2 + 1} \right) = (x, y, z)$$

The stereographic projection maps great circles to

- straight lines through the origin, if the great circle intersects the north pole (projection point)
- or circles otherwise

Proof. Let a great circle be given by its unit normal vector $\hat{n} = (n_x, n_y, n_z)$. Points on the great circle are given by

$$xn_x + yn_y + zn_z = 0$$

we substitute x, y, z by unprotecting the corresponding point on the plane (X, Y)

$$\Leftrightarrow n_x \frac{2X}{X^2 + Y^2 + 1} + n_y \frac{2Y}{X^2 + Y^2 + 1} + n_z \frac{X^2 + Y^2 - 1}{X^2 + Y^2 + 1} = 0$$

$$\Leftrightarrow 2Xn_x + 2Yn_y + n_z(X^2 + Y^2 - 1) = 0$$

If $n_z = 0$, in which case the great circle must intersect the north and south pole, the equation degenerates into a linear equation, giving us points on a straight line through the origin.

$$\Leftrightarrow 2Xn_x + 2Yn_y = 0$$

If $n \neq 0$ we divide by n_z

$$\Leftrightarrow 2X \frac{n_x}{n_z} + 2Y \frac{n_y}{n_z} + X^2 + Y^2 - 1 = 0$$

We substitute $a = 2 \frac{n_x}{n_z}$ and $b = 2 \frac{n_y}{n_z}$ for readability and rearrange to

$$\Leftrightarrow (X^2 + aX) + (Y^2 + bY) = 1$$

finally, by completing the square we end up with the equation of a circle

$$\Leftrightarrow \left(X + \frac{a}{2}\right)^2 + \left(Y + \frac{b}{2}\right)^2 = 1 + \left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2$$

We can determine the exact properties the projected great circles

In the first case, the line given by the normal vector $\hat{n}' = \begin{pmatrix} n_x \\ n_y \end{pmatrix}$

$$2Xn_x + 2Yn_y = 0$$

$$\Leftrightarrow Xn_x + Yn_y = 0$$

$$\Leftrightarrow \begin{pmatrix} X \\ Y \end{pmatrix} \cdot \begin{pmatrix} n_x \\ n_y \end{pmatrix} = 0$$

In the second case, the circle can be described by its center $C = (c_x, c_y)$ and radius r

The radius is given by

$$r = 1 + \left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2 = 1 + \frac{n_x^2 + n_y^2}{n_z^2}$$

$$c_x = -\frac{a}{2} = -\frac{n_x}{n_z}$$

$$c_y = -\frac{b}{2} = -\frac{n_y}{n_z}$$

- The circle is smallest with radius $r = 1$ when $\hat{n} = (0, 0, \pm 1)$, i.e. when the great circle lies on the equator
- The radius grows to infinity as $r \xrightarrow{n_z \rightarrow 0} \infty$
- ☐ show how the formula changes if we offset the sphere along the z-axis, while still projecting from it's north pole
- ☐ no isometric mapping
- ☐ to get a better understanding of hyperbolic models in the next chapter, we want to understand the projection S2 onto E2
- ☐ Great circles map to circles or lines
- ☐ Reflection on lines becomes inversion
- ☐ Excursion into mapping of transformations on the sphere
- ☐ Distortions through the projection

4.1 Polar Coordinates

- ☐ r : radius
- ☐ θ : inclination
- ☐ ϕ : azimuth (polar angle)
- ☐ conversion cartesian coordinates

4.2 Distance Metric

The distance between two points on a sphere is given by the minor arc length of a great circle.

- ☐ exercise generalise the formula for a sphere with radius R
- ☐ how we measure distance on the sphere
- ☐ how we measure distance in our model
- ☐ area metric
- ☐ central angle $\Delta\sigma = \mathbf{a} \cdot \mathbf{b}$

4.3 Area of a circle

Let P be a point on the sphere. We define a circle with radius r to be the set of points that are distance r away from P

The central angle is given by $\Delta\sigma = \frac{R}{r}$.

5 Standart hyperbolic plane

- ☐ example
- ☐ infinitelz many tesselations
- ☐ more space

5.1 Hyperbolic Models

Hyperbolic geometry is by itself much harder to grasp.

- ☐ Projection of paraboloid

5.2 Distance Metric

A

5.3 Transformations